STATISTICAL MODELING OF PARTICULATE MATTER ($PM_{2.5}$) IN NAIROBI

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DECLARATION

We declare that this research proposal is our original work and has not been presented elsewhere.

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LIST OF ABBREVIATIONS

$PM_{2.5}$	Particulate Matter of Diameter ≤ 2.5 micrometers
WHO	World Health Organization
DALYs	Disability-Adjusted Life Years
ARIMAX	Auto-Regressive Integrated Moving Average with Exogenous Variables
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
ADF	Augmented Dickey–Fuller test

Chapter 1: Introduction

1.1 Background of the Study

Air pollution is currently being cited as one of the gravest environmental hazards to the general population of the globe. According to the estimates of the World Health Organization (WHO), air pollution in the ambient (outdoor) environment claimed around 4.2 million premature lives in 2019, the number of which is disproportionately higher in low- and middle-income countries (World Health Organization, 2023). Of the many pollutants, the most hazardous one is the fine particulate matter of less than 2.5 micrometers ($PM_{2.5}$) in diameter, which can be absorbed deep in the lung passageways and into the blood and potentially cause cardiovascular, cerebral and respiratory effects. Recent Global Burden of Disease estimates indicate that the ($PM_{2.5}$) exposure has led to approximately 7.83 million deaths and more than 231 million Disability-Adjusted Life Years (DALYs) worldwide (Fang, et al., 2025).

In Sub-Saharan context, urbanization and industrialization have been so high as compared to the formulation of environmental regulations. Kenyan cities such as Nairobi are also witnessing concomitant increase in the density of population, motor traffic and unchecked industrial emissions. This has resulted into a situation whereby people are exposed to dangerous air pollution regularly. The annual mean ($PM_{2.5}$) concentration in Nairobi has been measured at about $18 - 19\ \mu g/m^3$, nearly four times the WHO's recommended annual guideline of $5\ \mu g/m^3$, and long-term exposure at this level is estimated to contribute to approximately $400 - 1{,}400$ premature deaths annually in the city's population. Nonetheless, this is a serious problem, but Nairobi does not have a large-scale, state-managed system of reference-quality air quality monitors. This information vacuum prevents the policymakers to introduce timely interventions, as well as citizens to implement preventive steps during episodes of high pollution levels. Nonetheless, the emergence of so-called citizen science and the use of inexpensive sensors (the PMS5003 sensors discussed herein) provide a decent alternative to data collection. The sensors give high-frequency information about PM, temperature, and humidity.

The problem, though, does not lie with the lack of data, but rather with the complexity of data. Raw sensor data is not usually smooth and is influenced by meteorological factors and cannot easily be interpreted without statistical processing. To convert this crude data into usable information on the health of the people, it is necessary to subject it to rigorous statistical modeling. In this project, it is proposed to model and predict ($PM_{2.5}$) behaviors in the city of Nairobi by using the Auto-Regressive Integrated Moving Average with Exogenous Variables (ARIMAX) with the help of the meteorological information adopted as major predictors.

1.2 Problem Statement

Nairobi experiences persistently elevated ($PM_{2.5}$) concentrations due to rapid urbanization, increasing traffic emissions, and limited regulatory enforcement, posing a significant public health risk. Although low-cost sensor networks and open secondary datasets now provide high-frequency air quality and meteorological data, these data are underutilized for systematic statistical modeling and forecasting (Kirago, Gatari, Gustafsson, & Andersson, 2022). The absence of validated, locally appropriate predictive models limits the ability of policymakers and health authorities to anticipate pollution episodes and design proactive air-quality management strategies.

1.3 Purpose of the Study

The primary purpose of this study is to develop and validate statistical models to analyze and forecast ($PM_{2.5}$) pollution levels in Nairobi using sensor-based time-series data. By transforming raw environmental data into predictive insights, the study aims to demonstrate the viability of using low-cost sensors for rigorous air quality monitoring in resource-constrained settings. The project aims to provide a methodological framework that allows for the prediction of air quality exceedances based on historical trends and meteorological forecasts.

1.4 Objectives of the Study

1.4.1 General Objective

To model and forecast ($PM_{2.5}$) concentrations in Nairobi using ARIMAX models that incorporate meteorological predictors.

1.4.2 Specific Objectives

1. To describe historical ($PM_{2.5}$) patterns in Nairobi, identifying seasonal, weekly, and daily fluctuations using sensor data.
2. To create ARIMAX models that include meteorological predictors (humidity and temperature) to quantify their influence on air quality.
3. To evaluate the accuracy of the developed models using Mean Absolute Error (MAE) and Root Mean Square Error (RMSE).
4. To forecast high-pollution episodes and identify specific days likely to exceed the WHO safety threshold of 25 $\mu g/m^3$.

1.5 Research Questions

1. What temporal trends characterize Nairobi's ($PM_{2.5}$) levels from 2017 to 2025?
2. Which time lags best represent the effects of temperature and humidity on ($PM_{2.5}$) concentrations?

3. Which ARIMAX model configurations yield the strongest forecasts in terms of error minimization?
4. What do short-term forecasts reveal about upcoming pollution episodes, and can they accurately predict exceedances?

1.6 Justification of the Study

The primary rationale for this research lies in its potential to inform critical public health and urban planning interventions. By statistically modeling the meteorological conditions that trigger pollution spikes, this study enables public health officials to accurately forecast and declare "Air Quality Action Days," thereby allowing vulnerable groups, such as children, the elderly, and asthmatics to take necessary precautions. Furthermore, the analysis of temporal trends provides evidence-based data for urban planners and policymakers to refine traffic management strategies. Specifically, demonstrating the statistical link between rush-hour traffic and peak pollution levels validates the need for specific interventions, such as congestion charges or improved public transportation systems, to mitigate environmental hazards.

Additionally, this study is justified by its contribution to the academic community through the validation of citizen-science data for rigorous research. In developing countries where official reference-grade monitoring infrastructure is often limited, this project demonstrates the methodological soundness of utilizing open-source secondary data for robust statistical modeling. It serves as a replicable template for other researchers, proving that accessible, low-cost data can be effectively transformed into predictive insights to bridge the information gap in environmental management.

1.7 Scope of the Study

The study is geographically confined to Nairobi City, specifically focusing on the urban monitoring site located at coordinates Lat -1.269, Lon 36.819. The research covers a longitudinal period of eight years, spanning from November 2017 to November 2025, to capture seasonal seasonality and long-term trends. Thematically, the scope is restricted to the statistical analysis of Fine Particulate Matter ($PM_{2.5}$) as the dependent variable, with Ambient Temperature and Relative Humidity included as the exogenous independent variables for the development of predictive models.

Methodologically, this is a quantitative desk study relying exclusively on secondary data retrieved from the *sensors.AFRICA* repository. The scope does not involve primary data collection, physical fieldwork, or laboratory analysis of the chemical composition of particulates. The analytical focus is limited to Time Series Analysis using ARIMA and ARIMAX models; therefore, other pollutants (such as NO_2 or SO_2) and direct clinical health assessments are excluded from this research.

Chapter 2: Literature Review

2.1 Health Impacts of Particulate Matter

Air pollution has become an area of concern in terms of its health burden in the urban African centers. Examined the issue of health consequences of fine particulate matter in Nairobi extensively (Oguge, et al., 2024). In their study, they have developed a definite statistical connection between ($PM_{2.5}$) exceedances over long periods of time and the rates of respiratory and cardiovascular mortality. They state that unless something is done, the public health infrastructure in Nairobi may not be able to meet the health bill related to the air pollution. Supported this trend worldwide as they examined the trend between 1990 and 2021 and expected that the burden of the disease would keep increasing in developing countries unless stringent control strategies are taken (Fang, et al., 2025).

2.2 Sources of Pollution in Nairobi

It is important to be able to model based on the sources of pollution. One of the earliest studies to measure traffic effects on air quality in Nairobi was conducted by (Kinney, et al., 2011), who found that certain hotspots in the city near large roadways of the city had very high levels of ($PM_{2.5}$) throughout the day. Extended this study by providing the analysis of black carbon and ascribing a significant part of the air pollution in Nairobi to the burning of fossil fuels in automobiles (Kirago, Gatari, Gustafsson, & Andersson, 2022). Their results indicate that the traffic patterns (morning and evening rush hours) are good predictors of pollution spikes, and this variable should be reflected in the statistical model by time-of-day analysis.

2.3 Role of Low-Cost Sensors in Urban Air Quality Monitoring

Without the government infrastructure, citizen science has taken their place. The sensors were used in informal settlements in Nairobi and to make the invisible visible, (West, et al., 2020) deployed low-cost sensors in these settlements. In their paper, they have shown that low-cost sensors (such as the PMS5003 utilized in the present project) have greater precision errors than reference monitors, but at the same time, they are very useful in tracing relative trends and detecting the episode of pollution when properly calibrated. Our proposed research methodology of utilizing the Open Africa dataset is confirmed by this research because (West, et al., 2020) demonstrated that community-based data collection has scientific importance as a tool of temporal analysis.

2.4 Statistical Modeling Approaches

Although the use of complex machine learning models (such as Neural Networks) is becoming popular, simple statistical approaches are still the best when it comes to interpretability. ARIMAX (Auto-Regressive Integrated Moving Average with Exogenous inputs) model is considered a solid tool in air quality prediction since it can cope with the autocorrelation of time-series data (today is predicted by yesterday pollution), whilst it is still able to include external regressors, such as weather. According to earlier research, humidity and temperature are some of the meteorological conditions that affect particles in significant but lagged ways and therefore, lag-distributed models are models that must be used (Kinney, et al., 2011).

2.5 Research Gap

Although the literature sets the sources (Kirago, Gatari, Gustafsson, & Andersson, 2022) and the health risks (Oguge, et al., 2024), there is a lack of literature that makes predictions at the daily level in Nairobi. The majority of studies that were conducted are retroactive (examining the old data). This is the gap that is filled in this project through the creation of a forward-looking predictive model.

3 **Chapter 3:** Methodology

3.1 Research Design

The research design used is quantitative, longitudinal, correlational. It is longitudinal in the way it examines data obtained over a considerable time (2017-2025), and correlational in that it is aimed at estimating the relationship between independent variables (Temperature, Humidity, Time) and the dependent variable ($PM_{2.5}$) without controlling the environment.

3.2 Target Population

In contrast to social science studies where human subjects are used, the population of this study includes high-frequency data points of the environment. The data is obtained in the Open Africa archive, namely, the Sensors.AFRICA project. The specific dataset covers:

- Geographic location: Nairobi, Kenya (Sensors at the sensor location of app Lat -1.269, Lon 36.819).
- Time Period: November 2017 to November 2025.
- Volume: There are thousands of rows of sensor readings (with time stamps) stored in the raw CSV file.

3.3 Instruments

This was done through cheap air quality devices. According to the provided data, the particular instruments are:

1. PMS503 Sensor: The sensor is a digital universal particle concentration sensor which is based on the laser scattering theory to determine the concentration of suspended particles in the air. It records PM_1, ($PM_{2.5}$) , and PM_{10}

2. DHT22 Sensor: A cheap, simple digital temperature and humidity sensor.
3. Data Processing Hardware: The sensors are normally attached to microcontrollers (such as NodeMCU) which are capable of transmitting data through Wi-Fi to the Open Africa repository.
4. Analysis Software: The data cleaning and statistical modeling will be done with Python (pandas, statsmodels, and scikit-learn libraries) or R by the research team.

3.4 Procedure

The following are the procedural steps that will be followed in the study:

1. Data Acquisition: The CSV file is obtained in the Open Africa repository.
2. Data Cleaning: The raw data is of mixed types and may have errors (e.g., NULL values in location columns). The rows containing missing target variables ($PM_{2.5}$) will be eliminated.
3. Resampling: The sensors are used to collect data which is not evenly spaced (e.g., every 1030 seconds). The data will be resampled to hourly averages and daily averages to reduce noise as well as make the data computationally amenable.
4. Stationarity Testing: Stationarity of the time series will be tested using Augmented Dickey-Fuller (ADF) test. In case the data is non-stationary (as it happens to environmental data), differencing will be used.
5. Model Fitting: 80 percent of the historical data will be used to train Linear regression and ARIMAX models.
6. Prediction: The models will make predictions of the other 20 percent of the data (test set).

3.5 Data Collection & Analysis

3.5.1 Data Structure:

The given data is of a long format. The sample row will be the following: 4930; pms5003;3981; −1.269;36.819;2025-10-01T00:00:16...; P2;28.40.

- P2 represents ($PM_{2.5}$).
- P1 represents PM_{10}
- P0 represents PM_1
- temperature and humidity are separate rows.

3.6 Analysis Plan:

1. **Pivoting:** The data will be reorganized (pivoted) to have the timestamps as an index and ($PM_{2.5}$), Temperature, and Humidity will be separate columns.
2. **Descriptive Statistics:** Mean, Median, Standard Deviation and Interquartile Range (IQR) are calculated to understand the distribution.
3. **ARIMAX Modeling:** The mathematical representation of a general form of the ARIMAX model to be used in this analysis is:

$$\boldsymbol{Y_t = \beta_0 + \beta_1 X_{1,t} + \beta_2 X_{2,t} + \varphi_1 Y_{t-1} + \ldots + \theta_1 \varepsilon_{t-1} + \varepsilon_t}$$

Where:

- $\boldsymbol{Y}_t$ is the ($PM_{2.5}$) concentration at time t.
- $\boldsymbol{\beta}_0$ is the constant (intercept).
- $\boldsymbol{X}_{1,t}$ and $\boldsymbol{X}_{2,t}$ are the exogenous variables (Temperature and Humidity) at time t.
- $\boldsymbol{\varphi}_1$ (Phi) represents the autoregressive coefficient (past values).
- $\boldsymbol{\theta}_1$ (Theta) represents the moving average coefficient (past errors).

- $\boldsymbol{\varepsilon}_t$ (Epsilon) is the white noise error term.

We will also use a baseline Linear Regression for comparison:

$$PM = \beta_0 + \beta_1 T + \beta_2 H + \varepsilon$$

4. **Metric Calculation:** The model's success will be determined by calculating:

- MAE (Mean Absolute Error): Mean size of errors.

$$MAE = \frac{1}{n}\sum_{t=1}^{n} | Y_t - \widehat{Y}_t |$$

- RMSE (Root Mean Square Error): The more significant the error, the more it is penalized.

$$RMSE = \sqrt{\frac{1}{n}\sum_{t=1}^{n} (Y_t - \widehat{Y}_t)^2}$$

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3.7 Work Plan

The proposed project duration is approximately 14 weeks.

ACTIVITIES	2025				2026			
	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr
Topic Identification & Proposal Writing			■					
Literature Review			■	■				
Methodology Development				■				
Proposal Submission & Defense				■	■			
Data Collection (Open Africa Downloads)					■			
Data Cleaning & Pre-processing					■			
Exploratory Data Analysis (Trends)						■		
Model Development (ARIMAX)						■	■	
Forecasting & Evaluation (RMSE/MAE)								■
Final Report Writing & Submission								■

3.8 Budget

As this project relies on secondary data available openly via the Internet, the direct costs are minimal.

Activity/ Item	Quantity	Unit Price (KES)	Total (KES)
High-Performance Laptop (Core i7)	1 Laptop	100,000	100,000
External Hard Drive (2TB for backup)	1 Unit	10,000	10,000
Internet Bundles (Large Dataset Downloads)	4 Months	3,000	12,000
Cloud Computing Credits (AWS/Colab Pro)	1 Subscription	15,000	15,000
Statistical Software License / Token	1 License	5,000	5,000
Stationery (Reams, Files, Pens)	Lump Sum		5,000
Printing Draft Proposals	5 Drafts	1,500	7,500
Typesetting & Formatting Services	60 Pages	100	6,000
Final Thesis Hard Binding	6 Copies	1,500	9,000
Airtime & Communication	6 Members	1,000	6,000
Subsistence / Meals during group work	Lump Sum	-	15,000
Journal Publication Fees	1 Paper	10,000	10,000
Contingency Fund (approx. 10%)			21,550
GRAND TOTAL			222,050

APPENDIX

The secondary data for this study will be extracted from the Sensors.AFRICA Air Quality Archive. The dataset is in long (narrow) format and contains high-frequency measurements indexed by timestamp and geolocation.

Description of Variables:

- sensor_id: Unique numeric identifier for each sensor (e.g., 4930).
- timestamp: Date and time of the measurement in ISO 8601 format.
- value_type: Type of measurement recorded. Possible values include:
- P2: $PM_{2.5}$ concentration (fine particulate matter ≤ 2.5 μm).
- P1: PM_{10} concentration (coarse particulate matter $\leq$ 10 μm).
- temperature: Ambient temperature in degrees Celsius ($°C$).
- humidity: Relative humidity in percentage (%).
- value: The numeric reading recorded by the sensor (μg/m^3 for PM, °C for temperature, % for humidity).
- lat / lon: Geographic coordinates of the sensor location in Nairobi.

Source: Sensors.AFRICA Air Quality Archive, Nairobi. Available at:

https://open.africa/dataset/sensorsafrica-airquality-archive-nairobi